

Name: \_\_\_\_\_ Teacher: \_\_\_\_\_

## YEAR 11 MATHEMATICS EXTENSION 1 ASSESSMENT TASK – JUNE 2009

**Time Allowed: 70 minutes**

**Full working should be shown in every question.**

**Marks may be deducted for careless or badly arranged work.**

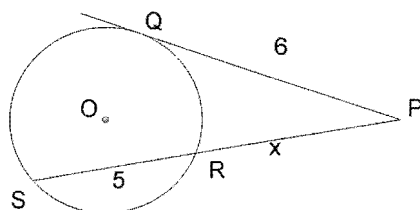
**No liquid paper is to be used.**

**If a correction is to be made, one line is to be ruled through the incorrect answer.**

**Marks**

1. PQ is a tangent to the circle below. Find  $x$  giving reasons where necessary.

3



2. Find the acute angle between the lines  $x - 2y + 1 = 0$  and  $x + 3y + 2 = 0$ .

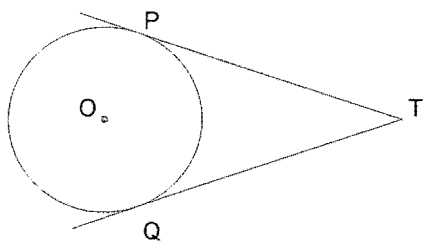
3

3. Solve  $2\tan\theta - 3\cot\theta = 5$  for  $0^\circ \leq \theta \leq 360^\circ$ .

3

4. PT and QT are tangents to a circle with centre O. Prove  $\angle PTQ = 2\angle PQO$

3



5. Express  $2\cos\theta + 2\sqrt{3}\sin\theta$  in the form  $R\cos(\theta - \alpha)$  and hence find the minimum value of the expression.

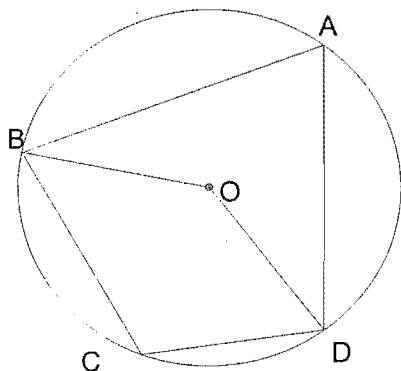
3

6. Prove, using  $t = \tan\frac{\theta}{2}$ , that  $\frac{1 + \operatorname{cosec}\theta}{\cot\theta} = \frac{1 + \tan\frac{\theta}{2}}{1 - \tan\frac{\theta}{2}}$

3

7. The point  $P(-2,9)$  divides the interval  $AB$  in the ratio  $k:1$ , where  $A$  is  $(7,-3)$  and  $B$  is  $(1,5)$ . Find the value of  $k$ . 2

8.  $\angle BAD = \alpha$  and  $\angle BOD = \angle BCD$ . Find  $\alpha$  giving reasons. 3



9. Prove  $\frac{\sin A}{\cos A + \sin A} + \frac{\sin A}{\cos A - \sin A} = \tan 2A$  3

10. The angle between two lines  $y = mx$  and  $y = \frac{1}{2}x$  is  $45^\circ$ . Find the possible values of  $m$ . 3

11. Solve  $\sqrt{2}\sin\theta - \cos\theta = 1$  for  $0^\circ \leq \theta \leq 360^\circ$  3

12.  $AB$  and  $CD$  are two intersecting chords of a circle and  $CD$  is parallel to the tangent to the circle at  $B$ .

(a) Draw a neat sketch of the above information.

- (b) Prove that  $AB$  bisects  $\angle CAD$ . 3

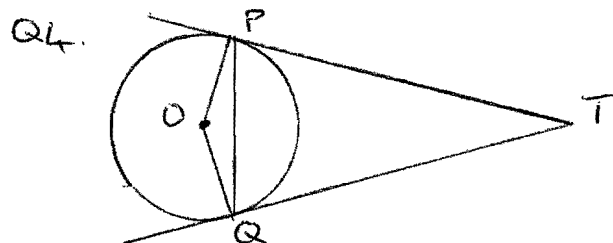
END OF TEST

Q1.  $x(x+9) = 6^2$  (sq. of tangent  
 $x^2 + 9x - 36 = 0$  = product of intercepts)  
 $(x+9)(x-4) = 0$   
 $x = -9$  or  $4$   
 since  $x > 0$ ,  $x = 4$  (3)

Q2  $x - 2y + 1 = 0 \therefore y = \frac{1}{2}x + \frac{1}{2} \therefore m_1 = \frac{1}{2}$   
 $x + 3y + 2 = 0 \therefore y = -\frac{1}{3}x - \frac{2}{3} \therefore m_2 = -\frac{1}{3}$   
 $\tan \alpha = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$   
 $= \left| \frac{\frac{1}{2} - (-\frac{1}{3})}{1 + \frac{1}{2} \times (-\frac{1}{3})} \right|$   
 $= \left| \frac{\frac{5}{6}}{\frac{5}{6}} \right|$   
 $= 1$  (3)

$\therefore \alpha = 45^\circ$   
 Q3  $2 \tan \theta - \frac{3}{\tan \theta} = 5$   
 $2 \tan^2 \theta - 3 = 5 \tan \theta$   
 $2 \tan^2 \theta - 5 \tan \theta - 3 = 0$   
 $2 \tan^2 \theta - 6 \tan \theta + \tan \theta - 3 = 0$   
 $2 \tan \theta (\tan \theta - 3) + 1(\tan \theta - 3) = 0$   
 $(2 \tan \theta + 1)(\tan \theta - 3) = 0$   
 $\therefore \tan \theta = -\frac{1}{2}$  or  $\tan \theta = 3$

$\therefore \theta = 153^\circ 26', 333^\circ 26', 71^\circ 34', 251^\circ 34'$   
 In order:  
 $\theta = 71^\circ 34', 153^\circ 26', 251^\circ 34', 333^\circ 26'$  (3)



Let  $\angle PQO = \alpha$   
 $\therefore \angle PQT = 90^\circ - \alpha$  ( $\angle$  bet. tang & radius =  $90^\circ$ )  
 $\Delta PQT$  is isosceles (tang. from ext. pt. are equal)  
 $\therefore \angle QPT = 90^\circ - \alpha$  (eq.  $\angle$ 's of isos.  $\Delta$ )  
 $\therefore \angle PTQ = 180^\circ - 2(90^\circ - \alpha)$  ( $\angle$  sum of  $\Delta$ )  
 $= 2\alpha$   
 $\therefore \angle PTQ = 2 \times \angle PQO$  (3)

Q5.  $2 \cos \theta + 2\sqrt{3} \sin \theta = R \cos(\theta - \alpha)$   
 $= R \cos \theta \cos \alpha + R \sin \theta \sin \alpha$   
 $\therefore R \cos \alpha = 2, R \sin \alpha = 2\sqrt{3}$   
 $\therefore R = \sqrt{2^2 + (2\sqrt{3})^2} = 4$   
 and  $\tan \alpha = \frac{2\sqrt{3}}{2} = \sqrt{3} \therefore \alpha = 60^\circ$   
 $\therefore \text{Exp.} = 4 \cos(\theta - 60^\circ)$   
 Min. Value =  $-4$  (3)

Q6. If  $\tan \frac{\theta}{2} = \frac{t}{1}$    
 then  $\cos \theta = \frac{1}{\sin \theta}$   
 $= \frac{1}{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}$   
 $= \frac{1}{2 \cdot \frac{t}{\sqrt{1+t^2}} \cdot \frac{1}{\sqrt{1+t^2}}}$   
 $= \frac{1+t^2}{2t}$   
 and  $\cot \theta = \frac{1}{\tan \theta}$   
 $= \frac{1}{\frac{2 \tan \frac{\theta}{2}}{1 - \tan^2 \frac{\theta}{2}}}$   
 $= \frac{1-t^2}{2t}$   
 $\therefore \text{LHS} = 1 + \frac{1+t^2}{\frac{1-t^2}{2t}}$   
 $= \frac{2t + 1+t^2}{1-t^2}$   
 $= \frac{(1+t)^2}{(1+t)(1-t)}$   
 $= \frac{1+t}{1-t}$   
 $= \text{RHS}$  (3)

Q7. Using x-values and  $m=k, n=1$   
 $x_p = \frac{mx_2 + nx_1}{m+n}$   
 $-2 = \frac{k \times 1 + 1 \times 7}{k+1}$   
 $-2k - 2 = k + 7$   
 $-3k = 9$   
 $k = -3$  (2)

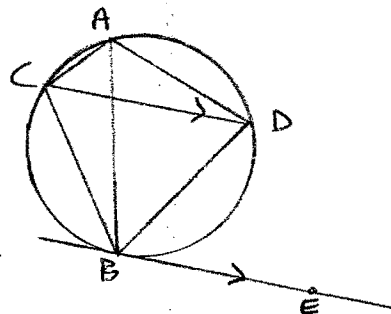
Q8.  $\angle BAD = x$  ✓ |  
 $\therefore \angle BOD = 2x$  ( $\angle$  at centre =  $2 \times \angle$  at circum. stand. on same arc)  
 $\angle BCD = 180^\circ - x$  (opp.  $\angle$ 's of a cyclic quad are suppl.)  
 But  $\angle BOD = \angle BCD$  (given) ✓ |  
 $2x = 180^\circ - x$   
 $3x = 180^\circ$   
 $x = 60^\circ$  — | (3)

Q9.  $LHS = \frac{\sin A \cos A - \sin^2 A + \cos A \sin A + \sin^2 A}{\cos^2 A - \sin^2 A}$  ✓ |  
 $= \frac{2 \sin A \cos A}{\cos^2 A - \sin^2 A}$   
 $= \frac{\sin 2A}{\cos 2A}$  — |  
 $= \tan 2A$   
 $= RHS$  (3)

Q10.  $\left| \frac{m - \frac{1}{2}}{1 + m + \frac{1}{2}} \right| = \tan 45^\circ = 1$  — |  
 $\frac{m - \frac{1}{2}}{1 + \frac{1}{2}m} = \pm 1$   
 $\therefore m - \frac{1}{2} = 1 + \frac{1}{2}m$  or  $m - \frac{1}{2} = -1 - \frac{1}{2}m$   
 $\frac{1}{2}m = \frac{3}{2}$  ✓ |  $\frac{3}{2}m = -\frac{1}{2}$  ✓ |  
 $m = 3$  or  $m = -\frac{1}{3}$  (3)

Q11. let  $\sqrt{2} \sin \theta - \cos \theta = R \sin(\theta - \alpha)$   
 $= R \sin \theta \cos \alpha - R \cos \theta \sin \alpha$   
 $\therefore R \cos \alpha = \sqrt{2}$ ,  $R \sin \alpha = 1$   
 $\therefore R = \sqrt{2+1} = \sqrt{3}$   
 and  $\tan \alpha = \frac{1}{\sqrt{2}} \therefore \alpha = 35^\circ 16'$  ✓ |  
 $\therefore \sqrt{3} \sin(\theta - 35^\circ 16') = 1$   
 $\sin(\theta - 35^\circ 16') = \frac{1}{\sqrt{3}}$  — |  
 $\therefore \theta - 35^\circ 16' = 35^\circ 16', 144^\circ 44'$   
 $\therefore \theta = 70^\circ 32', 180^\circ$  — | (3)

Q12. (a)



(b) Draw AC, AD, BC, BD  
 let  $\angle DBE = \alpha$  ✓ |  
 $\therefore \angle BAD = \alpha$  ( $\angle$  bet. tang & chord =  $\angle$  in altern. seg.)  
 $\angle BDC = \alpha$  (Altern.  $\angle$ 's,  $CD \parallel BE$ )  
 $\angle BAC = \alpha$  ( $\angle$ 's on circum stand on the same arc BC)  
 $\therefore \angle BAD = \angle BAC$  (3)

Alternative Q11 soln. ✓

If  $\sqrt{2} \sin \theta - \cos \theta = 1$   
 let  $t = \tan \frac{\theta}{2}$   
 $\therefore \sin \theta = \frac{2t}{1+t^2}$ ,  $\cos \theta = \frac{1-t^2}{1+t^2}$   
 $\therefore \sqrt{2} \times \frac{2t}{1+t^2} - \frac{1-t^2}{1+t^2} = 1$  — |  
 $2\sqrt{2}t - 1 + t^2 = 1 + t^2$   
 $2\sqrt{2}t = 2$   
 $\tan \frac{\theta}{2} = \frac{1}{\sqrt{2}}$   
 $\therefore \frac{\theta}{2} = 35^\circ 16'$   
 $\theta = 70^\circ 32'$  — |

Check  $\theta = 180^\circ$  — |  
 $LHS = \sqrt{2} \times 0 - 1 = -1 = RHS$   
 $\therefore \theta = 70^\circ 32', 180^\circ$  (3)